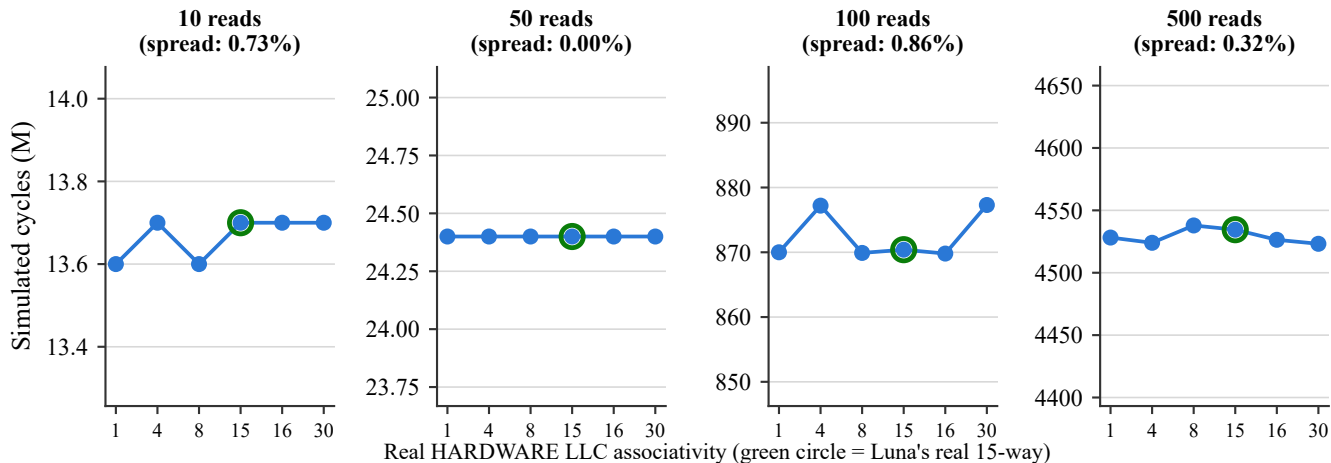


Hardware LLC associativity alone has no measurable effect (base kraken2, NO software cache, 50MB DB)



Software S2 cache held fixed at NONE throughout (base kraken2 binary) - the inverse of every other experiment in this project, which varied the software cache instead. Y-axis is zoomed to each panel's own tiny range to show there is no trend - all four panels show <1% spread with no monotonic direction. NULL RESULT: at this 50MB DB, almost nothing is ever resolved at the LLC layer (l2/nuca hit rates all <2%), so its associativity has nothing to act on - the same underlying reason the software cache also fails to help on this DB (see fig9/fig10/fig11).